

CONCEPT CHECK 42.1

1. How is the flow of hemolymph through an open circulatory system similar to the flow of water through an outdoor fountain?
2. Three-chambered hearts with incomplete septa were once viewed as being less adapted to circulatory function than mammalian hearts. What advantage of such hearts did this viewpoint overlook?
3. **WHAT IF?** The heart of a normally developing human fetus has a hole between the left and right atria. In some cases, this hole does not close completely before birth. If the hole weren't surgically corrected, how would it affect the O_2 content of the blood entering the systemic circuit?

For suggested answers, see Appendix A.

CONCEPT 42.2

Coordinated cycles of heart contraction drive double circulation in mammals

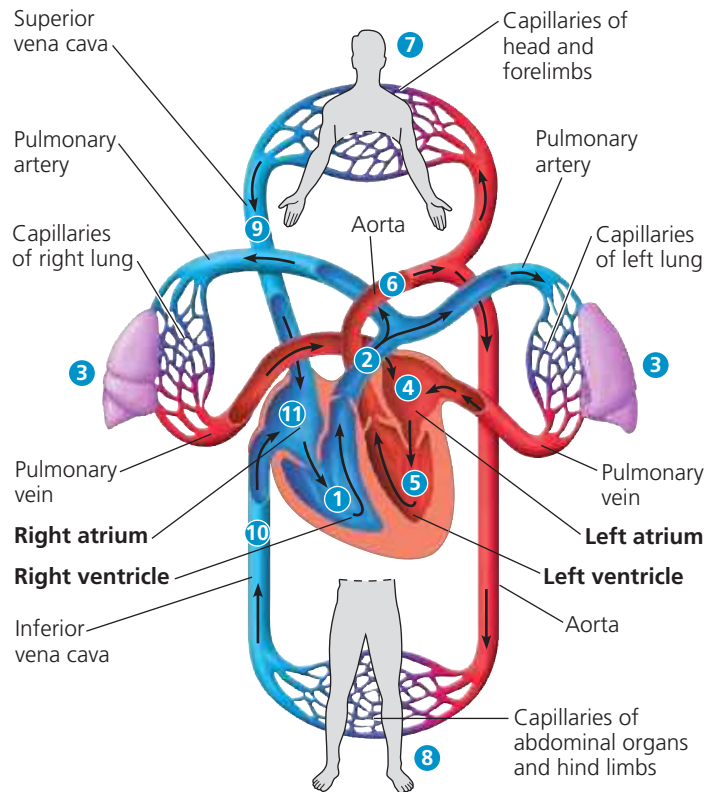
The timely delivery of O_2 to the body's organs is critical. Some brain cells, for example, die if their O_2 supply is interrupted for even a few minutes. How does the mammalian cardiovascular system meet the body's continuous (although variable) demand for O_2 ? To answer this question, we must consider how the parts of the system are arranged and how each of these parts functions.

Mammalian Circulation

Let's first examine the overall organization of the mammalian cardiovascular system, beginning with the pulmonary circuit. (The circled numbers refer to labeled structures in **Figure 42.5**.) Contraction of **1** the right ventricle pumps blood to the lungs via **2** the pulmonary arteries. As the blood flows through **3** capillary beds in the left and right lungs, it loads O_2 and unloads CO_2 . Oxygen-rich blood returns from the lungs via the pulmonary veins to **4** the left atrium of the heart. Next, the oxygen-rich blood flows into **5** the heart's left ventricle, which pumps the oxygen-rich blood out to body tissues through the systemic circuit. Blood leaves the left ventricle via **6** the aorta, which conveys blood to arteries leading throughout the body. The first branches leading from the aorta are the coronary arteries (not shown), which supply blood to the heart muscle itself. Branches further along the aorta lead to **7** capillary beds in the head and arms (forelimbs). The aorta then descends into the abdomen, supplying oxygen-rich blood to arteries leading to **8** capillary beds in the abdominal organs and legs (hind limbs). Within the capillary beds, there is a net diffusion of O_2 from the blood to the tissues and of CO_2 (produced by cellular respiration) into the blood. Capillaries rejoin, forming venules, which convey blood to veins. Oxygen-poor blood from the head, neck, and

▼ Figure 42.5 The mammalian cardiovascular system:

an overview. Note that the dual circuits operate simultaneously, not in the serial fashion that the numbering in the diagram suggests. The two ventricles contract almost in unison and pump the same volume of blood. However, the total volume of blood in the systemic circuit is much greater than that in the pulmonary circuit.



VISUAL SKILLS If you trace the path of a molecule of carbon dioxide that starts in an arteriole in the right thumb and leaves the body in exhaled air, what is the minimum number of capillary beds the molecule encounters? Explain.

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forelimbs is channeled into a large vein, **9** the superior vena cava. Another large vein, **10** the inferior vena cava, drains blood from the trunk and hind limbs. The two venae cavae empty their blood into **11** the right atrium, from which the oxygen-poor blood flows into the right ventricle.

The Mammalian Heart: A Closer Look

Using the human heart as an example, let's now take a closer look at how the mammalian heart works (**Figure 42.6**). Located behind the sternum (breastbone), the human heart is about the size of a clenched fist and consists mostly of cardiac muscle (see Figure 40.5). The two atria have relatively thin walls and serve as collection chambers for blood returning to the heart from the lungs or other body tissues. Much of the blood that enters the atria flows into the ventricles while all four heart chambers are relaxed. The remainder is transferred by contraction of the atria before the ventricles